Assignment 2

Exponential family form

~~$f(x|\theta) = h(x) \exp(\eta(\theta) \cdot T(x) - A(\theta))$~~

$$P(y;\eta) = b(y) \exp(\eta T(y) - a(\eta)) \quad - ①$$

$T(y)$ = sufficient stat $[(T(y) = y)] \rightarrow$ Same dimension as η

$b(y)$ - base measure - not dependant on η

$a(\eta)$ = log partition function

y, a, b are scalar func.

$$P(y|\lambda) = \frac{\lambda^{y} \cdot e^{-\lambda}}{y!}$$

$$= \exp\left(\log\left(\frac{\lambda^{y} \cdot e^{-\lambda}}{y!}\right)\right)$$

$$= \exp(y \log\lambda - \lambda - \log(y!))$$

$$= \frac{1}{y!} \exp(y \log\lambda - \lambda)$$

Comparing with ①

$T(y) = y \qquad a = e^{\eta}$

$\eta = \log y \qquad b(y) = \frac{1}{y!}$

~~Mean: $E(T(y))$~~ $T(y)$ is k-1 dimension vector

For binary logistic regression

$y \in (0,1) \qquad T(y) = y$

Using ~~encoding~~

$y \qquad T(y)$

$1 \qquad (1,0)$

2

$y \in \{1, 2 \cdots k\}$

$$T_i(y) = \begin{cases} 1 & y = i \\ 0 & \text{otherwise} \end{cases}$$

then $T(y) = (T_1(y), T_2(y) \cdots T_{(k-1)}(y))$

The last class k is a reference class

Since if

$p_1 + p_2 + p_3 = 1$

$p_3 = 1 - p_1 - p_2$ so we need on k-1 parameters (independant)

So η_i is defined as

$\eta_i = \log \frac{p_i}{p_k}$

Defining probabilities

$p_i = P(y=i)$ $\quad p_1 = P(Y=1)$

$p_2 = P(Y=2)$

$p_i \geq 0 \quad \sum_{i=1}^{k} p_i = 1$

In exponential form, we first write distribution

$P(Y=y) = \prod_{i=1}^{k} p_i^{1(y=i)}$ $\quad$ Eg: if $y=2$ $\quad p_1^0 p_2^1 p_3^0 = p_2$

$$\log P(Y=y) = \sum_{i=1}^{k-1} 1(y=i) \log p_i + 1(y=k) \log p_k$$

writing $\log p_i = \log \frac{p_i}{p_k} + \log p_k$

$$= \sum_{i=1}^{k-1} 1(y=i) \log \frac{p_i}{p_k} + \log p_k$$

Since $\eta_i = \frac{p_i}{p_k}$

& $T_i(y) = 1(y=i)$

$$= \sum_{i=1}^{k-1} T_i(y) \eta_i + \log p_k$$

$$P(y) = \exp\left(\sum_{i=1}^{k-1} T_i(y)\eta_i + \log p_k\right)$$

$$a(\eta) = -\log p_k$$

$$\eta_i = \log \frac{p_i}{p_k}$$

$e^{\eta_i} = \frac{p_i}{p_k} \Rightarrow p_i = p_k e^{\eta_i}$

$\sum_{i=1}^{k} p_i = 1$ (Prob. condition)

$\Rightarrow p_k + \sum_{i=1}^{k-1} p_k e^{\eta_i} = 1$

$\Rightarrow p_k \left(1 + \sum e^{\eta_i}\right) = 1$

$\therefore a(\eta) = \log\left(1 + \sum_{i=1}^{k-1} e^{\eta_i}\right) \quad p_k = \frac{1}{1 + \sum_{i=1}^{k-1} e^{\eta_i}}$

Mean $\quad \mu_i = \frac{\partial A(\eta)}{\partial \eta_i} = P(Y=k)$ $\quad [\because T_i(y) = 1$ when $y=i)$

$$= \frac{e^{\eta_i}}{1 + \sum_{j=1}^{k-1} e^{\eta_j}}$$

$E[T_i(Y)] = \sum_y T_i(y) P(Y=y)$

GLM

$$\eta_i = x^T \beta_i$$

β_i → parameter

x → feature

$$P(Y = i \mid x) = \frac{e^{x^T \beta_i}}{1 + \sum_{j=1}^{k-1} e^{x^T \beta_j}}$$